Problem solving using C

30 October 2026 20:36

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hour/30)
		L	T	P	L	T	P	SL	NL	
1111104	Problem Solving Using C	02	--	--	30	--	--	30	60	2

Course Code	Course Name	Theory					TW	PR / OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
1111104	Problem Solving Using C	20	20	40	60	2	---	---	100

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

The course is designed in alignment with the principles of NEP 2020 with emphasis on skill-based, experiential, multidisciplinary, and industry-oriented learning. Aligned with the Computer Engineering theme "Digital India", the course introduces students to the fundamentals of C programming with focus on logical reasoning, algorithm development, and problem-solving skills. Students will learn structured and modular programming concepts using control structures, arrays, functions, pointers, structures, and file handling.

The curriculum integrates real-world applications such as banking systems, healthcare utilities, inventory management, smart billing systems, and digital record management to promote practical and application-based learning. Emphasis is also given to coding standards, debugging techniques, and secure programming practices used in modern software industries. The course supports the vision of Digital India by developing computational and technology-driven problem-solving abilities for future engineering applications. It also contributes toward UNSDG 4 (Quality Education), UNSDG 8 (Decent Work and Economic Growth), UNSDG 9 (Industry, Innovation and Infrastructure), UNSDG 11 (Sustainable Cities and Communities), and UNSDG 16 (Peace, Justice and Strong Institutions).

Course Objectives:

Sr. No.	Course Objectives
1	To introduce students to the fundamentals of programming using C language.
2	To develop logical thinking and problem-solving skills.
3	To enable students to write simple and structured C programs.
4	To familiarize students with decision making, loops, arrays, functions, and pointers.
5	To develop confidence in beginners for coding and debugging
6	To prepare students for advanced programming and software development courses.

Course Outcomes: On successful completion of the course, the learner will be able to:

Sr. No.	Course Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Understand the basic structure and syntax of the C programming language.	L2
2	Develop C programs using operators, conditional statements, and loops.	L3
3	Implement programs using arrays and strings	L3
4	Design modular programs using functions for code reusability, parameter passing, and problem solving.	L6
5	Apply pointers and structures in problem solving.	L3
6	Perform file handling operations in C.	L3

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
0	Prerequisites	Introduction to computers, software and hardware, programming languages, algorithm and flowchart, basics of logic building, introduction to compiler and execution of program.		-
1	Computational Thinking and C Fundamentals	**1.1 Introduction to Problem Solving and Computational Thinking:** Problem Solving Concepts, Computational Thinking Basics, Algorithms, Flowcharts, Pseudocode **1.2 Fundamentals of C Programming:** Overview and Features of C Language, Structure of a C Program, Compilation and Execution Process, Tokens, Keywords, Identifiers, Variables, Constants, Data Types, Operators, Input-Output Functions **Self-Learning Topics:** Write-up on history and evolution of programming languages , Practicing Non-Formatted Inp& Output functions – getchar( ), putchar( ), getch( ), getche( ), putch( ), gets( ), puts( ) , Practicing Formatted Input & Output functions – printf( ) **Online References:** https://www.programiz.com/c-programming , https://www.geeksforgeeks.org/c-programming-language/	04	CO_01
2	Decision Making and Iterative Problem Solving	**2.1 Conditional statements** :if, if-else, nested if, switch-case, **2.2 Iterative statements** :for, while, do-while, nested loops, jump statements, menu-driven programming and problem solving using control structures. **Self-Learning:** Pattern programs (Pyramids , Star , Square , Upper triangular matrix , Lower triangular matrix) **Online References:**	06	CO_02

		https://www.tutorialspoint.com/cprogramming/c_decision_making.htm , https://www.programiz.com/c-programming/c-for-loop		
3	Arrays and Strings	**3.1 Introduction to Arrays** : Declaration ,definition ,accessing array elements,One-dimensional and two-dimensional arrays, matrix operations,array counting. **3.2 Strings:** Declaration, Initialization, String Operations, Array of Strings, String manipulation Functions in string.h **Self-Learning:** Sorting an array, Binary searching , Declaration and Initialization of a Multidimensional Array , Arrays of strings: Two-dimensional character array , Passing Arrays to Functions **Online References:** https://www.geeksforgeeks.org/arrays-in-c-cpp/ , https://www.programiz.com/c-programming/c-arrays	06	CO_03
4	Modular Programming with Functions	**4.1 Function:** Defining a Function, accessing a Function, Passing Arguments to a Function, call by value,call by Reference ,Recursion **4.2 Storage classes** : auto,static,register,extern. **Self-Learning:** Implement recursive solutions (e.g., Fibonacci, Tower of Hanoi) , Comparison of recursion and iteration (mini presentation) , Industry application: Use of arrays/strings in data processing. **Online References:** https://www.geeksforgeeks.org/functions-in-c/ , s	06	CO_04
5	Pointers and Structure	**5.1 Pointers:** Basics, address operator (&), Declaring & initializing Pointer Variables, Indirection Operator and Dereferencing, Pointer Expressions and Pointer Arithmetic, Types of Pointers, Pointer to Pointers, Pointers and Arrays, Call. by value vs Call by address,malloc, calloc, realloc, free. **5.2 Structures and Unions** :Declaring Structures and Structure Variables, Accessing the members of a structure, Initialization of Structures, typedef, Nested Structures, Arrays of Structures **Self-Learning:** Call by address using pointers, returning more than one value from a function, returning pointer from a function, structures and pointers, passing structure to function ,Structure versus Union, Enumeration types , Processing binary files	05	CO_05

		Online References: https://www.programiz.com/c-programming/c-pointer **nptel.ac.in/courses/106104128**		
6	File Handling	**6.1 Basic File Operations**: Opening, reading, writing, and closing a .txt file using fopen(), fscanf(), fprintf(), and fclose(). **Self-Learning:** Understand the purpose of file handling in C programming,Learn the difference between text files and binary files,File modes, **Online References:** https://www.programiz.com/c-programming/c-file-input-output https://www.geeksforgeeks.org/basics-file-handling-c/	03	CO_06
			Total Hours	**30**

Note:

1. Students are encouraged to explore self-learning topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination.
2. **Course Engagement (ARCS-based):** The engagement component aims to motivate active student participation in classroom learning. At the end of selected lectures, students will provide short reflective inputs—such as concepts they feel confident about or aspects that brought clarity or satisfaction.

References:

1. E. Balagurusamy, "Programming in ANSI C", McGraw Hill Education, 8th Edition / 2019.
2. Brian W. Kernighan and Dennis M. Ritchie, "The C Programming Language", Pearson Education, 2nd Edition / 2015.
3. Yashavant Kanetkar, "Let Us C", BPB Publications, 18th Edition / 2021.
4. Reema Thareja, "Programming in C", Oxford University Press, 2nd Edition / 2016.

Online References:

Sr. No.	Website Name	URL
1	Coursera – C Programming Specialization	https://www.coursera.org/specializations/c programming
2	NPTEL – Programming in C	https://onlinecourses.nptel.ac.in/e learning/preview/noc19_cs42
3	Udemy – C Programming Courses	https://www.udemy.com/topic/c-programming

Assessment:

Internal Assessment Test (IAT) for 20 marks each: IA will consist of Two Compulsory Internal Assessment Tests. Approximately 40% to 50% of the syllabus content must be covered in the IAT-I, and the remaining 40% to 50% of the syllabus content must be covered in the IAT-II.

Self Learning (SL): Self-learning is integrated into both theory and practical subjects, particularly where practical sessions have only one contact hour per week. It emphasizes student-driven efforts in preparing for assignments, practicals, and examinations. Additionally, self-learning includes topics that go beyond the prescribed syllabus to align with current industry trends and requirements. Assessment of self-learning will follow a formative approach. For theory subjects that include term work, 30% of the assignment questions and student presentations/discussions should be based on self-learning topics. In subjects without term work, the corresponding laboratory sessions will incorporate self-learning, carrying a weightage of 30%**.** This may involve lab work using the latest tools or preparing technical write-ups/research papers on advanced topics.

End Semester Theory Examination:

Question paper format:

- Question Paper will comprise a **total of six questions** each carrying 15 marks. Q.1 will be compulsory and should cover the maximum contents of the syllabus.
- Remaining questions will be mixed in nature (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)
- A **total of four questions** need to be answered.